

You are the senior full-stack engineer for PSM OS.

Task: Build the PTW Permit Detail Page after the Create New Permit wizard.

Route:

- /ptw/:id

Goal: When a user opens a permit, every visible card, tab, lifecycle button, counter, badge, table, side panel, and preview must load real API data, save changes to database, and refresh instantly.

Do NOT create a static demo page. Do NOT use mock cards, fake counters, or hardcoded permit data. Do NOT redesign the mockup. Keep the current dark industrial UI style.

Build backend first, then frontend.

Tech stack:

- Backend: NestJS + Prisma/Supabase + PostgreSQL + Redis/BullMQ
- Frontend: Next.js 14 + TypeScript + Tailwind + shadcn/ui + React Query + React Hook Form + Zod
- Storage: S3/Supabase Storage
- PDF: existing PDF engine
- Realtime: Supabase Realtime/WebSocket where available

PTW Detail Page must be site-aware:

- Every query must enforce company_id and site_id.
- User can only open permits from assigned sites.
- Backend must enforce access, not only frontend.

===== PAGE LAYOUT
=====

Follow the attached PTW mockup layout:

Left:

- PTW sidebar/list
- Search permits
- Status filters
- Active/Pending/Expiring/History tabs
- Permit cards
- Pagination

Center:

- Permit header

- Lifecycle stepper
- Action buttons
- Permit tabs
- Main tab content cards

Right:

- Permit summary
- Gas retest warning
- Area map / permit map
- Compliance / approvals
- Permit actions

===== MAIN HEADER
=====

Show real permit data:

- Permit number
- Permit title
- Permit type badge
- Status badge
- Risk level badge
- Equipment tag
- Area
- Permit holder
- Issued by
- Current lifecycle status
- Expiry countdown

Header action buttons:

- Submit
- Approve
- Issue Permit
- Activate
- Suspend
- Extend
- Close Permit
- Cancel
- Download Permit
- Print Permit
- Duplicate Permit
- Create MOC from Permit

Rules:

- Only show allowed actions based on status and user permission.

- Backend validates every lifecycle transition.
- No lifecycle step can be skipped.
- Every action creates audit log and permit history event.
- React Query must invalidate/refetch permit detail, dashboard counters, permit list, equipment linked records, and notifications.

===== LIFECYCLE STEPPER
=====

Statuses:

- Draft
- Submitted
- Approved
- Issued
- Active
- Suspended
- Extended
- Closed
- Cancelled

Stepper must show:

- Completed steps
- Current active step
- Blocked steps
- Timestamp per completed step
- User who completed step

Lifecycle rules:

- Draft → Submitted
- Submitted → Approved
- Approved → Issued
- Issued → Active
- Active → Suspended / Extended / Closed
- Suspended → Active after revalidation
- Active/Issued → Cancelled if permitted
- Closed is final

Hard blocks:

- Cannot activate if required approvals incomplete.
- Cannot activate if isolation incomplete.
- Cannot activate if gas test required but missing/failed.
- Cannot activate if required workforce briefing incomplete.
- Cannot activate if unresolved conflict exists.
- Cannot close if de-isolation incomplete.

- Cannot close if workforce sign-out incomplete.
- Cannot close without closure verification.

===== REQUIRED TABS
=====

Create functional tabs:

1. Details
2. Isolation
3. Gas Test
4. Workforce
5. Shift Handover
6. Conflicts
7. Signatures
8. History
9. Attachments

Each tab must:

- Load from API
- Show loading state
- Show empty state
- Show error state
- Allow permitted mutations
- Save to database
- Refresh instantly after mutation

===== 1. DETAILS TAB
=====

Show real permit information:

Permit Information:

- Permit number
- Permit type
- Permit status
- Risk level
- Priority
- Company
- Site
- Department
- Process unit
- Area
- Location description
- GPS latitude/longitude
- Planned start

- Planned end
- Issue date/time
- Expiry date/time
- Shift
- Permit holder
- Permit issuer
- Area authority
- Performing authority
- Contractor company

Work Description:

- Permit title
- Detailed work description
- Work method/procedure
- Tools and equipment required
- Chemicals/materials used
- Energy sources involved
- SOP reference
- P&ID reference
- Special instructions

Equipment Context:

- Equipment tag
- Equipment name
- Equipment type
- Equipment criticality
- Parent equipment
- Child equipment
- Linked equipment tags
- Active permits on same equipment
- Open MOCs/actions linked to equipment

Type-Specific Requirements: Show fields based on permit type:

- Hot Work
- Cold Work
- Confined Space
- Electrical Isolation / LOTO
- Excavation
- Radiography
- Working at Height
- Line Breaking / Equipment Opening
- SIMOPS

Actions:

- Edit permit details
- Save changes
- Refetch all cards after save

===== 2. ISOLATION TAB
=====

Build full isolation/LOTO management.

Show:

- Isolation required
- Isolation completion percentage
- Isolation authority
- Isolation certificate status
- De-isolation status
- Isolation points table

Isolation point fields:

- Energy source
- Energy type
- Isolation point tag
- Valve tag / breaker tag
- Required position
- Current position
- Lock number
- Lock holder
- Blind/spade number
- Installed by
- Confirmed by
- Confirmed timestamp
- Verification required
- Second-person verification
- Notes
- Status

Actions:

- Add isolation point
- Edit isolation point
- Delete isolation point
- Confirm isolation
- Verify isolation
- Start de-isolation
- Confirm de-isolation

- Generate isolation certificate PDF

Rules:

- Permit cannot activate until all required isolations are confirmed.
- If second-person verification is required, confirmation alone is not enough.
- De-isolation must be complete before closure.
- Every confirmation stores user, timestamp, signature, and IP.
- Every mutation creates permit history and audit log.

===== 3. GAS TEST TAB
=====

Build full gas testing workflow.

Show:

- Gas test required
- Latest gas readings
- Pass/fail status
- Tester
- Instrument ID
- Instrument calibration date
- Last test timestamp
- Next retest due
- Retest countdown
- Gas test history table
- Gas trend chart if possible

Gas fields:

- O2
- LEL
- H2S
- CO
- SO2
- Cl2
- NH3
- HF
- Custom gases

Threshold rules:

- O2 valid range: 19.5–23.5%.
- Confined Space LEL default: <10%.
- Hot Work LEL default: 0% preferred.
- Hot Work site override may allow up to <5% only if site policy allows.
- H2S, CO, and custom gases are site-configurable.

- Failed gas test blocks activation.
- Failed gas test on active permit suspends permit.
- Retest overdue auto-suspends permit.

Actions:

- Add gas test
- Edit gas test if permitted
- View gas test history
- Configure retest interval if permitted

===== 4. WORKFORCE TAB
=====

Build workforce/personnel on permit.

Show:

- Permit holder
- Performing authority
- Workers table
- Contractor company
- Maximum personnel count
- Briefing completion
- Sign-in/sign-out status
- Emergency accountability list

Worker fields:

- Name
- Company
- Contractor company
- Trade
- Badge ID
- Contact number
- Briefing required
- Briefing completed
- Sign-in time
- Sign-out time
- Emergency contact
- Status

Actions:

- Add worker
- Edit worker
- Remove worker
- Mark briefing completed

- Sign in worker
- Sign out worker

Rules:

- Permit cannot activate if required briefing is incomplete.
- Permit cannot close if workers are still signed in.
- Sign-in/out stores user, timestamp, and IP.
- Every mutation creates permit history and audit log.

===== 5. SHIFT HANDOVER TAB
=====

Build digital shift handover.

Show:

- Active permit summary
- Current shift
- Outgoing supervisor
- Incoming supervisor
- Handover time
- Permit status at handover
- Permit risks
- Gas test validity
- Isolation validity
- Permit expiry countdown
- Open conflicts
- Permits expiring within 2 hours
- Handover checklist
- Incoming supervisor acknowledgement
- Handover signature
- Timestamp
- IP address

Checklist:

- Reviewed work scope
- Reviewed isolation status
- Reviewed gas test status
- Reviewed open conflicts
- Reviewed expiring permits
- Accepted responsibility for permit continuation

Actions:

- Create handover
- Acknowledge handover

- Suspend permit during handover

Rules:

- Incoming supervisor must acknowledge active permit.
- Handover acknowledgement is an electronic signature.
- Handover creates audit log and permit history event.

===== 6. CONFLICTS TAB
=====

Build conflict detection and override.

Show conflicts:

- Same equipment conflict
- Same area conflict
- Hot Work + Confined Space conflict
- Hot Work + Line Breaking conflict
- SIMOPS conflict
- Incompatible permit matrix conflict
- Nearby active permits

Conflict fields:

- Conflict type
- Severity
- Description
- Related permit
- Related equipment
- Area/unit
- Detected at
- Status
- Override status
- Override justification
- Overridden by
- Override timestamp

Actions:

- Run conflict check
- View related permit
- Override conflict if permitted

Rules:

- Conflict blocks activation unless overridden.
- Override requires Area Authority permission and written justification.

- Notify Permit Issuer, Area Authority, and Control Room.
- Override is permanently audit-logged.

===== 7. SIGNATURES TAB =====

Build electronic signature workflow.

Required signatures depend on permit type.

Possible signatures:

- Permit Holder
- Permit Issuer
- Area Authority
- Performing Authority
- Gas Tester
- Isolation Authority
- HSE
- Fire Watch
- Incoming Supervisor
- Closure Authority

Signature fields:

- Role
- Required user/role
- Signed by
- Status
- Timestamp
- IP address
- Comment

Actions:

- Sign
- Reject signature step if permitted
- View signature history

Rules:

- User must authenticate before signing.
- Signature cannot be forged by another user.
- Permit cannot progress if required signatures incomplete.
- All signatures create audit logs and history events.

===== 8. HISTORY TAB =====

Show full permit timeline.

Events:

- Permit created
- Details edited
- Submitted
- Approved
- Issued
- Activated
- Suspended
- Extended
- Closed
- Cancelled
- Isolation added
- Isolation confirmed
- De-isolation completed
- Gas test added
- Gas test failed
- Gas retest overdue
- Worker added
- Worker signed in
- Worker signed out
- Conflict detected
- Conflict overridden
- Handover completed
- Signature completed
- Attachment uploaded
- Certificate generated

Each event shows:

- Date/time
- User
- Event type
- Description
- Before value
- After value
- Related record link

===== 9. ATTACHMENTS TAB
=====

Support permit files.

Attachment types:

- Job Safety Analysis

- Method Statement
- Rescue Plan
- Excavation Drawing
- Radiography Plan
- Electrical Isolation Drawing
- P&ID
- SOP
- LOTO Photo
- Gas Test Certificate
- Worksite Photo
- Closure Photo
- Other

Features:

- Upload
- Preview
- Download
- Delete
- Link existing Document Control document
- Show uploaded by
- Show uploaded date

Rules:

- Confined Space requires Rescue Plan.
- Excavation may require Excavation Drawing.
- Radiography requires Radiography Plan.
- High/Critical risk requires JSA or Method Statement.
- Upload/delete refreshes instantly.
- Upload/delete creates audit log and history event.

===== RIGHT SIDE PANEL
=====

Permit Summary:

- Active permits count
- Expiring soon count
- Suspended count
- Open conflicts count

Gas Retest Card:

- Show next retest due
- Countdown
- Overdue warning
- Auto-suspended indicator

Area Map:

- Active permit markers
- Equipment marker
- Conflict markers
- Expiring permit markers
- If SVG map is not available, create data-driven area list placeholder.

Compliance / Approvals:

- Approval completion percentage
- Required signatures status
- Isolation completion
- Gas test status
- Workforce briefing status

Permit Actions:

- Download Permit
- Print Permit
- Duplicate Permit
- Create MOC from Permit
- Generate Isolation Certificate

===== LEFT PERMIT LIST
=====

Build real sidebar permit register:

Features:

- Search
- Filter by Active
- Filter by Pending
- Filter by Expiring
- Filter by History
- Filter by permit type
- Filter by risk
- Permit cards
- Pagination

Each card shows:

- Permit number
- Title
- Area
- Holder
- Status

- Permit type
- Expiry countdown

Data must come from API.

===== BACKEND REQUIREMENTS
=====

Create/update APIs:

- GET /ptw/:id
- PATCH /ptw/:id
- GET /ptw/:id/summary
- GET /ptw/:id/isolation
- POST /ptw/:id/isolation
- PATCH /ptw/:id/isolation/:isolationId
- DELETE /ptw/:id/isolation/:isolationId
- PATCH /ptw/:id/isolation/:isolationId/confirm
- PATCH /ptw/:id/isolation/:isolationId/verify
- POST /ptw/:id/de-isolation
- GET /ptw/:id/gas-tests
- POST /ptw/:id/gas-tests
- PATCH /ptw/:id/gas-tests/:gasTestId
- GET /ptw/:id/workforce
- POST /ptw/:id/workforce
- PATCH /ptw/:id/workforce/:workerId
- DELETE /ptw/:id/workforce/:workerId
- POST /ptw/:id/workforce/:workerId/sign-in
- POST /ptw/:id/workforce/:workerId/sign-out
- GET /ptw/:id/handover
- POST /ptw/:id/handover
- POST /ptw/:id/handover/:handoverId/acknowledge
- GET /ptw/:id/conflicts
- POST /ptw/:id/conflicts/check
- POST /ptw/:id/conflicts/:conflictId/override
- GET /ptw/:id/signatures
- POST /ptw/:id/signatures
- GET /ptw/:id/history
- GET /ptw/:id/attachments
- POST /ptw/:id/attachments
- DELETE /ptw/:id/attachments/:attachmentId
- POST /ptw/:id/submit
- POST /ptw/:id/approve
- POST /ptw/:id/issue
- POST /ptw/:id/activate
- POST /ptw/:id/suspend
- POST /ptw/:id/extend
- POST /ptw/:id/close

- POST /ptw/:id/cancel
- GET /ptw/:id/certificate
- GET /ptw/:id/isolation-certificate

===== FRONTEND FILES
=====

Create/fix:

- app/(app)/ptw/[id]/page.tsx
- features/ptw/components/PermitDetailPage.tsx
- features/ptw/components/PermitHeader.tsx
- features/ptw/components/PermitLifecycleStepper.tsx
- features/ptw/components/PermitSidebarList.tsx
- features/ptw/components/PermitRightPanel.tsx
- features/ptw/components/tabs/PermitDetailsTab.tsx
- features/ptw/components/tabs/PermitIsolationTab.tsx
- features/ptw/components/tabs/PermitGasTestTab.tsx
- features/ptw/components/tabs/PermitWorkforceTab.tsx
- features/ptw/components/tabs/PermitShiftHandoverTab.tsx
- features/ptw/components/tabs/PermitConflictsTab.tsx
- features/ptw/components/tabs/PermitSignaturesTab.tsx
- features/ptw/components/tabs/PermitHistoryTab.tsx
- features/ptw/components/tabs/PermitAttachmentsTab.tsx
- features/ptw/hooks/usePermitDetail.ts
- features/ptw/hooks/usePermitMutations.ts
- features/ptw/services/ptw.service.ts
- features/ptw/schemas/permit-detail.schema.ts

===== ACCEPTANCE CRITERIA
=====

Permit Detail Page is complete only when:

- ✓ /ptw/:id loads real permit data
- ✓ Header shows real data
- ✓ Lifecycle stepper works
- ✓ Action buttons work based on status/permission
- ✓ Details tab works
- ✓ Isolation tab works
- ✓ Gas Test tab works

- ✓ Workforce tab works
- ✓ Shift Handover tab works
- ✓ Conflicts tab works
- ✓ Signatures tab works
- ✓ History tab works
- ✓ Attachments tab works
- ✓ Right summary panel works
- ✓ Left permit list works
- ✓ All counters are real
- ✓ All uploads refresh instantly
- ✓ All mutations save to database
- ✓ React Query invalidates after every mutation
- ✓ Backend validates lifecycle transitions
- ✓ Audit logs are created
- ✓ Permit history events are created
- ✓ No static/mock data remains

Final output: 1. Files created 2. Files modified 3. Backend changes 4. Frontend changes 5. API endpoints added 6. Tabs completed 7. Bugs fixed 8. Remaining gaps, if any